

Yeowon Yoon

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EDUCATION

Troy High School | Student

Aug 2022 – Present

- **11th GPA:** 3.95 unweighted, 4.72 weighted
- **AP Scores:** AP Computer Science Principles (5) - Perfect Score Recognition, AP Computer Science A (5), AP Calculus BC (5), AP Statistics (5), AP Physics 1 (5), AP Physics 2 (5), AP Physics C Mechanics (5), AP Physics C Electricity and Magnetism (5), AP Chemistry (5), AP Biology(5), AP World History (5), AP US History (5)
- **SAT:** 1550 (Math: 800)
- **ACT:** 34
- **Dual enrollment @ CSUF:** Math 250A-Calculus III, Math 250B-Linear Algebra and Differential Equations Math 335: Mathematical probability, Math 438-Stochastic Process, Math 320-Mathematical Computation

PREPROFESSIONAL EXPERIENCES

Machine Learning Troy Tech Intern

June 2025

Prof. Joshua Tallman, Concordia University Irvine

- Built a full-stack machine learning model for network intrusion detection using NetFlow data (NF-UNSW-NB15).
- Engineered features and trained a Random Forest classifier (99% accuracy, 91% recall) alongside a Feedforward Neural Network.
- Deployed results through an interactive web dashboard; project adopted into Professor Tallman's curriculum for instruction.

MIT Beaver Works Medyletics Program

July 2025

MIT Lincoln Laboratory – Beaver Works Summer Institute

- Participating in a 4-week program focused on AI in medicine and healthcare applications.
- Developing a final ML model using Python, NumPy, TensorFlow, and Git.
- Coursework includes predictive modeling, data preprocessing, and evaluation metrics.
- Developed app NeuroPlay, developing FNN and RF classifier to detect Parkinson's disease through an interactive game that detects biomarkers of neurodegenerative diseases
- Co-authored and published a peer-reviewed paper at the MIT Undergraduate Research Technology Conference (URTC), one of the nation's premier venues for undergraduate-led research in engineering and computation—accepted as a high school student.

COSMOS – Cluster 5: Computers in Biophysics & Robotics

June – July 2024

University of California, Davis

- Simulated bacterial run-and-tumble motion across a Gaussian chemoattractant gradient using the diffusion equation and Python libraries (NumPy, Matplotlib).
- Applied stochastic modeling and random walk theory to describe molecular behavior in biological systems.
- Constructed and programmed autonomous Lego MindStorm™ robots using C and EV3 sensors for final project
- Presented final projects in an Ignite-style presentation to university faculty and researchers.

Beaver Works Online Certificate – Python Core, Medyletics, Version Control

April 2023

MIT Beaver Works Pre-Program Curriculum

Completed three technical tracks:

- **Python Core:** Intermediate syntax, functions, recursion, data structures, and error handling
- **Medyletics:** AI applications in health data, model building, regression/classification using NumPy and TensorFlow
- **Version Control:** Git/GitHub fundamentals for collaborative, reproducible data science workflows

UCLA Summer Public Speaking & Debate Program

July – August 2022

University of California, Los Angeles

- Refined rhetorical strategies and argument structures in structured debate formats.
- Practiced 2–3 debates weekly, using flow notes and timed rebuttal planning to improve real-time speaking performance.

University of California, Santa Barbara

- Collaborated with student leaders in policy-centered workshops.
- Developed and proposed community initiatives through group discussion, goal-setting, and scenario-based planning.

LEADERSHIP AND COMMUNITY EXPERIENCES

Web Development Club

August 2023 – Present

Co-Founder & President | Troy High School

- Led weekly HTML/CSS/JavaScript workshops and facilitated project planning during board meetings.
- Oversaw development of 10+ live websites for school organizations including Machine Learning Club and OC Teen Inspiration.
- Partnered with 3&ME Academy to expand computer science education, contributing over 120 verified volunteer hours.
- Launched the official Troy High School Clubs Directory (troywebdev.com), the first centralized digital hub for student organizations, improving accessibility and streamlining engagement across the entire school community. 40 Clubs currently registered.
- Selected as 8 clubs to represent tech program Troy Tech fair

Physics Club

August 2025 – Present

Founder & President | Troy High School

- Established the school's Physics Club and expanded STEM outreach through competition and demonstration.
- Led team to 4th place in Southern California PhysicsBowl; served as top scorer on school team.
- Hosted the $F=MA$ exam and internal TroPhy competitions; delivered engaging physics demos to peers.

Coding Club

February 2025 – Present

President

- Designed curriculum covering Python fundamentals, data structures, and algorithmic thinking.
- Prepared coding activities and slides, and led biweekly hands-on educational meetings.
- Mentored new members during practice sessions and fostered collaborative peer learning.

Math Club

September 2024 – Present

Co-Founder & President | Matematika

- Co-authored a competition-prep textbook covering advanced math problems (AMC/HMMT/MathCon).
- Organized problem-solving workshops and led student teams in national math competitions.
- Promoted creative and logical reasoning through peer-based discussion and challenge sets.

GYM (Global Youth Mission)

June 2023 – Present

General Cabinet Member

- Helped plan monthly discussions and service projects addressing global social issues.
- Collaborated with peers to organize volunteer initiatives, educational workshops, and awareness campaigns.
- Contributed to community service efforts focused on underserved populations.

STEM Club

August 2022 – Present

General Cabinet Member

- Collaborated with the leadership team to design monthly STEM themes such as robotics, environmental science, and biotechnology.
- Prepared educational presentations and facilitated interactive science demonstrations during meetings.
- Engaged members through hands-on activities to promote interest in STEM fields.

Discovery Cube Team

April 2025 – Present

Team Staff Lead | Discovery Cube Orange County

- Supervised and mentored new and group volunteers during large-scale programs like Bubblefest and Winter Lights.
- Coordinated check-in/out, shirt distribution, shift schedules, and break coverage.
- Liaised between volunteers and museum management to maintain high-quality guest experiences.
- Acknowledged as a DCOC recognized volunteer.

- Assisted in a translation project to make exhibits more accessible to visitors.

Game Development Club

January 2025 – Present

Outreach Director & Lead Graphic Designer | Troy High School

- Directed school-wide outreach initiatives to grow club membership and foster external partnerships.
- Designed professional-grade digital assets for club events, games, and competitions.
- Collaborated with other executive board members on branding and public engagement strategy.

EXTRACURRICULAR ACTIVITIES

Marching Band / Concert Band

August 2022 – Present

1st Chair Symphonic Band | Troy High School

- Performed as 1st chair instrumentalist in Concert Band, demonstrating musical excellence and leadership.
- Participated in regional parades, field shows, school concerts, and competitive marching band events.
- Contributed to ensemble success through rehearsals, sectional leadership, and solo performances.

School Golf Team

November 2023 – Present

JV Player | Troy High School

- Competed in Spring 2024 and 2025 golf seasons against regional schools.
- Focused on refining swing technique, course strategy, and mental composure under tournament pressure.

Aim Wonder

October 2024 – Present

Volunteer | Global Humanitarian Organization

- Donated and produced inspirational videos for communities affected by the Israeli-Palestinian conflict.
- Advocated for vulnerable populations, including women in Palestine and Romani families, through multimedia outreach.

Taekwondo

Early Childhood – Present

Martial Artist | Private Dojang

- Earned a 3rd Degree Black Belt through years of disciplined training and testing.
- Demonstrated long-term commitment, physical conditioning, and leadership through peer mentoring.

Schoolhouse SAT Tutoring

January 2024 – Present

Volunteer SAT Tutor | College Board x Schoolhouse.world

- Selected based on top-tier SAT Reading performance to lead online tutoring sessions.
- Taught up to 20 students per session, focusing on evidence-based strategies and test pacing.

AWARDS AND DISTINCTIONS

USAPhO Qualified | AAPT – Physics, Feb 2024

NMSC NHSPC 2nd Place (National) | National High School Physics Competition – Physics, Apr 2025

NMSC NHSPC 2nd Place (California) | National High School Physics Competition – Physics, Apr 2025

PhysicsBowl School Top Scorer & Team 4th Place | AAPT – Physics (Southern California Region), Apr 2025

Peer-Reviewed Publication Primary Author | MIT Undergraduate Research Technology Conference – Research Paper (National), Oct 2025

AIME Qualified (AMC 12 A&B) | Mathematical Association of America – Mathematics, Nov 2024

MathCon Finalist (2X) | MathCon – Mathematics, 2023, 2024

Beaver Works Certificate | MIT Beaver Works – Computer Science & Machine Learning, Apr 2023

Presidential Volunteer Service Award – Gold | Discovery Cube – Community Service, Apr 2023

Concert Band 1st Chair Performer | Troy High School – Music Performance, 2023–2025

Congressional Award (Gold Medal, Expected) | The Congressional Award Foundation – National Service, 2026

AP Scholar with Distinction (2X) | College Board – Academics, July 2024 & July 2025

Science Olympiad State 5th Place (Mechatronics) | Science Olympiad – Engineering, Grade 8

M3 Challenge Top 20% | Society for Industrial and Applied Mathematics – Applied Mathematics, 2024

OCSEF Outstanding Achievement | Orange County Science & Engineering Fair – Science Research, March 2025

Discovery Cube Recognized Volunteer | Discovery Cube Orange County – Community Service, 2025